Questions .

Global q wrongly rejects all genuine Class B samples — they sit left of it

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PART 1 — Global vs. class-conditional conformal thresholds

Why does standard split conformal use one global threshold?

The guarantee in Proposition 3 comes from exchangeability of a *single* sequence: $\{s_1, ..., s_n, s(X)\}$. The proof is purely combinatorial — under exchangeability, the rank of $s(X)$ among $n+1$ values is uniform on $\{1, ..., n+1\}$, so $P(\text{rank} > k) \leq \alpha$ regardless of what the scores actually measure. This argument makes no reference to class identity at all. The "class" of a sample is just metadata the exchangeability proof never touches. That's precisely why it's called *marginal* coverage — it controls $P(s(X) > q)$ averaged over the entire known-class distribution, not $P(s(X) > q \mid \text{class} = c)$ for each c separately. One pooled threshold is the natural object because the pooled calibration set is exchangeable with a pooled test stream; nothing in Prop 3 as stated gives you per-class control.

Why does a global threshold create over/under-confidence across classes?

This is exactly your example. Say Class A prototype is very tight (fidelities cluster 0.95–0.98) because its samples are geometrically compact in Hilbert space — low intra-class variance, near-orthogonal to everything else. Class B's prototype is loose (fidelities 0.80–0.83) — maybe the class itself has more internal diversity, or its samples sit closer to the decision boundary with other classes.

A global q is computed from the *pooled* nonconformity scores $s(x) = 1 - F_{max}(x)$ across both classes. If Class A dominates the calibration set numerically (as your RSTFIN/SYN/TCP situation suggests can happen), the empirical quantile q will be pulled toward Class A's tight distribution — call it $q \approx 0.05$ (i.e. $F_{max} \approx 0.95$ cutoff). Now:

- **Class A samples**: comfortably clear q — this class is *over-served*, effectively getting a looser real false-alarm rate than the nominal α (its 5th-percentile calibration point is far from its bulk, wasting margin).
- **Class B samples**: a genuine, non-adversarial Class B sample at $F_{max} = 0.81$ has $s(x) = 0.19 > q = 0.05$ — it gets **rejected as a false zero-day**, even though nothing is wrong with it. This class is *under-served* — its real false-alarm rate is inflated far above the nominal α, because the threshold was set by a different class's geometry.

The *marginal* guarantee ($P(s(X) > q) \leq \alpha$ averaged over the whole known distribution) can still hold in aggregate even while Class B's own false-alarm rate is wildly higher than α and Class A's is wildly lower — the errors cancel across classes, not within any one of them. This is the formal statement of "unfair": Proposition 3 as written never promised class-wise fairness, only average fairness.

What is Class-Conditional (Mondrian) Conformal Prediction?

Vovk's Mondrian conformal prediction generalizes split conformal by partitioning the exchangeable sequence into disjoint groups (here: classes) and running the *same* conformal machinery independently within each group. For each class c, define the calibration subset $\{s(x) : x \in D_{cal}, y=c\}$ of size n_c, and compute:

$q_c = k_c$-th smallest value in that subset, $k_c = \lceil(1-\alpha)(n_c+1)\rceil$

Then accept x (predicted class c) iff $s_c(x) \leq q_c$. Everything about Prop 3's proof still applies — it's just re-run on K separate exchangeable sequences instead of one. The exchangeability requirement becomes **within-class exchangeability**: calibration samples of class c must be exchangeable with test samples of class c, which is actually an easier assumption to satisfy than global exchangeability (you don't need cross-class distributional stability, just within-class).

Is it mathematically valid?

Yes, unconditionally — Mondrian conformal is a standard, fully rigorous construction (Vovk, Gammerman & Shafer, *Algorithmic Learning in a Random World*). Each class gets its own valid marginal guarantee:

$P(s_c(X) > q_c \mid Y=c) \le \alpha$ for every c

Summing/averaging these over classes (weighted by class prevalence) recovers something *at least as strong as* the pooled marginal guarantee — in fact strictly stronger, because it holds conditionally on class, which implies the marginal version but not vice versa.

What assumption changes?

The single global assumption "calibration and test are exchangeable" becomes K separate assumptions "calibration-of-class-c and test-of-class-c are exchangeable," each checked independently (e.g. per-class AUROC discriminability tests). And critically — **the finite-sample size condition must now hold separately per class**: $n_c \ge 1/\alpha - 1$. This is exactly the constraint the theory PDF flags in Section 5.1 as the reason CICIoT2023 (with rare classes under 10 rows) can't use this approach globally. But note: this is a *per-dataset* empirical fact, not a property of Mondrian conformal itself. Your 6-class run has ~4000 calibration samples roughly evenly split, so each class almost certainly clears $n_c \ge 19$ at $\alpha=0.05$ comfortably.

Advantages / disadvantages

	Global (marginal)	Class-wise (Mondrian)
Guarantee strength	Weaker (averaged)	Stronger (per-class)
Fairness across classes	No — can silently starve one class	Yes, by construction
Data requirement	One pool of size $\ge 1/\alpha - 1$	Every class needs $\ge 1/\alpha - 1$
Behavior with rare classes	Degrades gracefully (borrows strength)	Must abstain per class or fall back
Complexity	Trivial	K calibration runs, K thresholds to report/maintain

Which should you prefer for zero-day detection?

Class-wise, whenever your data supports it — and the argument is stronger for a *security* application than a generic ML one: a detector that silently over-rejects one attack family (crying wolf on legitimate SYN-flood traffic) while under-rejecting another (letting borderline TCP-flood variants slip through at an inflated real false-alarm rate) is a worse security posture than one with uniformly-calibrated behavior per class, even if the pooled average looks fine on paper.

Given your per-class fidelity geometry is already visibly different (as your earlier RSTFIN/SYN/TCP results showed), a single global threshold is actively hiding the problem rather than reporting it honestly. Use the abstention-to-global-threshold fallback (from `min_calibration_size`) only for classes that genuinely can't support their own q_c.

All four Class B dots sit between q_B (their own, correct threshold) and the global q — meaning every one of those genuine samples gets misclassified as a false alarm under a pooled threshold, purely because Class A's tighter geometry dragged the global cutoff too high.

PART 2 — Why min and max? The worst-case logic of Fin and Fout

The mathematical role of `min` in `F_in`

Assumption (A2) states $F_{max}(x) \geq F_{in}$ for *every* known sample x. For this to be a valid statement about the whole class, F_{in} must be the **infimum** (in practice, minimum over the finite sample) — the single hardest-to-classify known point sets the bound for the entire guarantee. Concretely, from the theory doc's proof strategy step 1:

$d_{min}(x) \leq \sqrt{1 - F_{in}^2} =: d_{in}$ for **all** known x

This only holds as a *for-all* statement if F_{in} is a lower envelope on $F_{max}(x)$ — i.e. the min. Using anything larger (say the mean) would make the inequality false for every point below that mean.

The mathematical role of `max` in `F_out`

Symmetrically, (A3) states $F_{max}(z) \leq F_{out}$ for *every* zero-day sample z. For this to hold universally, F_{out} must be the **supremum** — the *closest-to-known* zero-day point defines the bound, because it's the hardest case to reject:

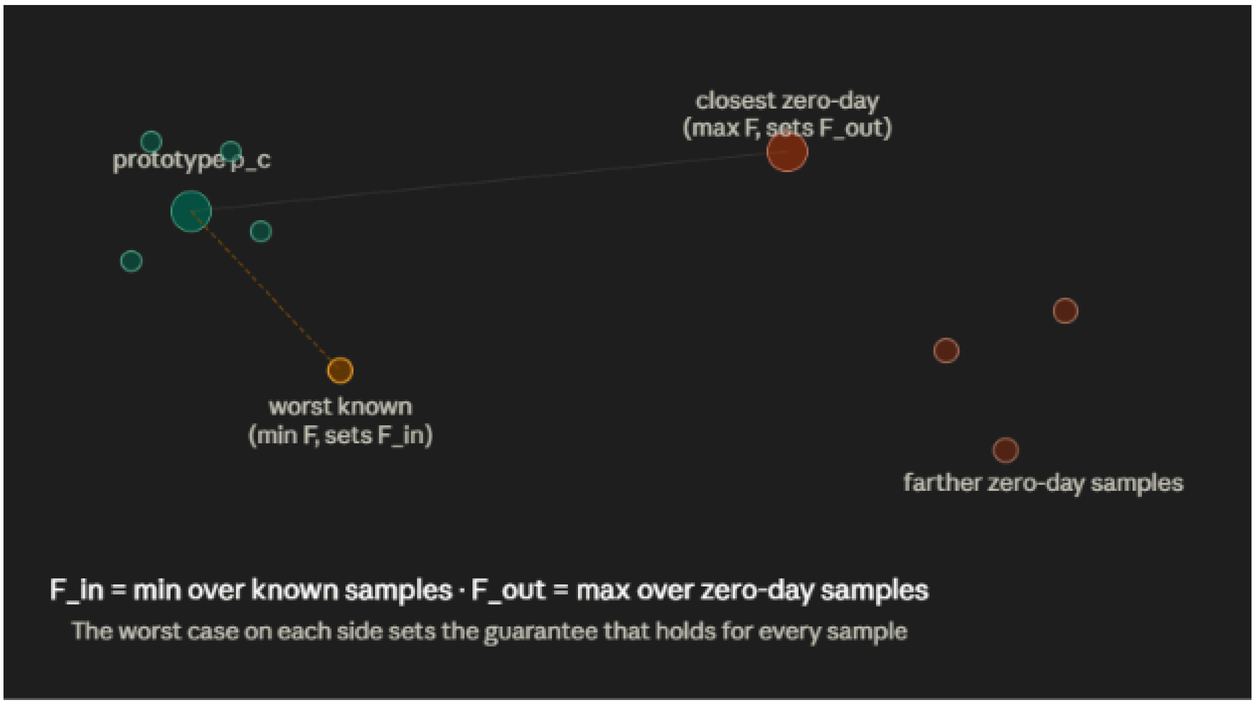
$d_{min}(z) \geq 1 - F_{out} =: d_{out}$ for **all** zero-day z

What guarantee do you get?

By taking the worst member on each side, you get a guarantee that holds for *every* known sample and *every* zero-day sample simultaneously — a genuine worst-case, distribution-free-in-form certificate (though not distribution-free in the Prop-3 sense; it's a Lipschitz/geometric argument, not a conformal one). This is what lets Prop 2 be a *theorem* rather than an average-case empirical claim: "provided $\varepsilon < \varepsilon^*$, **every** perturbed known sample is accepted and **every** zero-day sample is rejected" — no probability, no expectation, an absolute statement over the whole (assumed) support.

Why this makes the theorem "worst-case certified"

A certificate that only held for the *average* known sample would be worthless against an adversary — an attacker doesn't need to fool the average sample, just one. By anchoring `F_in` to the worst known sample and `F_out` to the worst zero-day sample, the resulting `Δ` and `ε*` are the largest gap/budget you can *guarantee* holds no matter which sample an adversary targets or which zero-day instance shows up. This is the same logic as Lipschitz-based certified robustness in classical ML (Cohen et al., randomized smoothing; Wong & Kolter, convex outer bounds) — certification is inherently a min-max object, and both `min` and `max` here are doing exactly the job a certified-robustness argument requires them to do.



The single-outlier fragility of Δ

Why does one outlier flip `F_in` to 0.20?

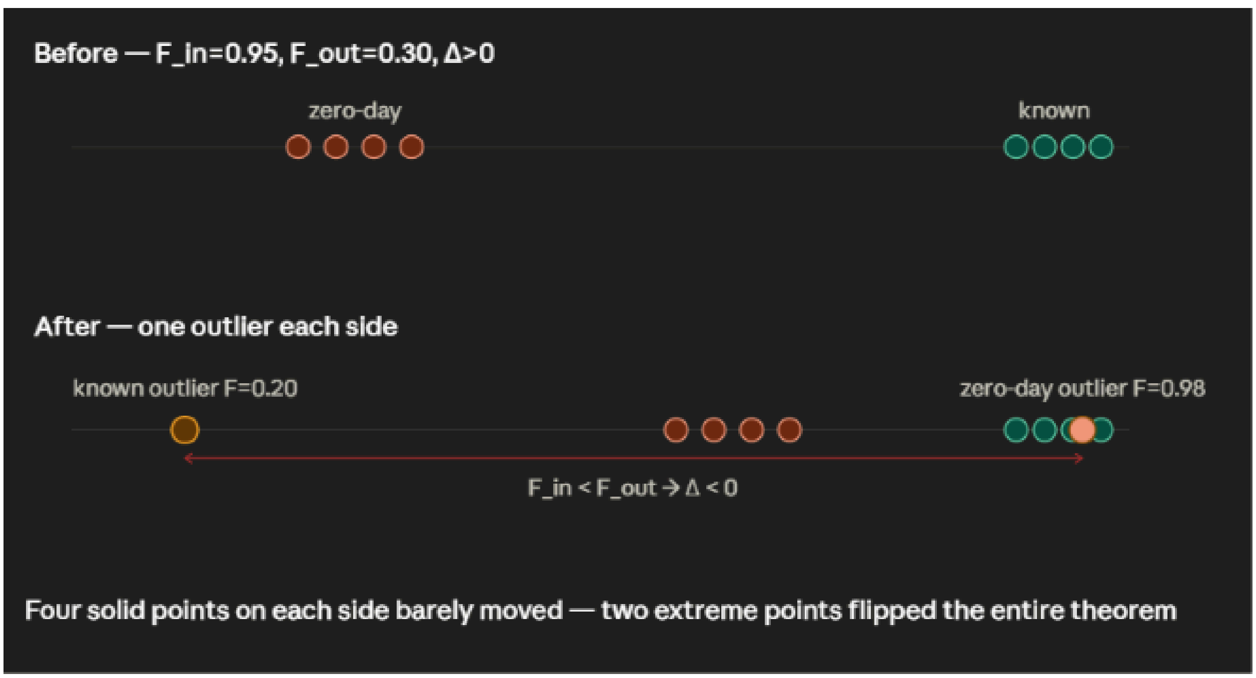
`F_in = min({0.98, 0.97, 0.96, 0.95, 0.20})`. `min` is the least robust statistic there is — a single point at the tail of the distribution *is* the statistic, regardless of how many points sit elsewhere. Four samples at 0.95+ contribute nothing to the value once one sample at 0.20 exists in the set. This is the textbook definition of a **non-robust estimator**: its breakdown point is `1/n` — one bad sample out of `n` is enough to make the estimator arbitrarily bad.

Why does one strange zero-day sample push `F_out` to 0.98?

Same mechanism, opposite direction: $F\_out = \max(\{0.15, 0.20, 0.25, 0.30, 0.98\})$. One zero-day instance that happens to sit unusually close to a known prototype (in your CICIoT2023/TON_IoT case: a genuinely novel attack that still shares feature-space structure with a known one — very plausible for closely-related DDoS variants) becomes the entire bound.

Why does Δ go negative?

$\Delta = (1 - F\_out) - \sqrt{1 - F\_in^2}$. Plug in: $(1 - 0.98) - \sqrt{1 - 0.20^2} = 0.02 - \sqrt{0.96} = 0.02 - 0.9798 \approx -0.9598$. Before the outliers (using the "clean" values, $F\_in \approx 0.95$, $F\_out \approx 0.30$): $\Delta = (1-0.30) - \sqrt{1-0.95^2} = 0.70 - \sqrt{0.0975} \approx 0.70 - 0.312 \approx 0.388 > 0$ — separable. **A single point on each side flipped the sign of the entire theorem's precondition**, from a comfortable positive margin to a deeply negative one. This is exactly the fragility the theory doc warns about in Section 3.5, and it's the mathematical reason worst-case guarantees in this framework are so brittle in practice — they inherit the breakdown point of `min`/`max`, which is the worst possible breakdown point any statistic can have.



Why your conformal calibration worked while Proposition 2 failed completely

Your numbers: $L_\varphi=1.5$, $F\_in=0.2015$, $F\_out=0.9790$, $\Delta=-0.9585$, $\varepsilon^*=-0.3227$, $q=0.4055$, with a known sample at $F\_max=0.8536$ correctly accepted. This is actually a very clean illustration of a conceptual point that's easy to blur together if you don't separate the two mechanisms carefully.

These are two structurally independent guarantees, answering different questions:

- **Proposition 3 (conformal)** answers: *"given the fidelity scores I actually observed on calibration data, what threshold makes the false-alarm rate on fresh known data ≤ α?"* It is a **purely statistical, data-adaptive** procedure. It doesn't care *why* the scores look the way they do, whether classes are separated, or whether any Lipschitz bound holds — it just computes an empirical quantile of whatever distribution of scores it's handed and proves a rank-statistics fact about it. This is why it worked: `q=0.4055` was computed *from your actual calibration score distribution*, whatever that distribution's shape, and the rank-based proof holds regardless of that shape.
- **Proposition 2 (certified separation)** answers a completely different, much harder question: *"is there a threshold that would remain valid even if an adversary perturbed a known input by up to ε?"* This requires an actual geometric guarantee — that known and zero-day samples occupy provably non-overlapping regions of fidelity space, wide enough that a bounded perturbation can't cross from one region to the other. It is **model-geometry-dependent**, not just data-adaptive: it needs the *worst* known sample to be far *in absolute distance* from the *closest* zero-day sample, which is a much stronger structural requirement than "there exists a good empirical quantile."

Why one can work while the other fails — geometrically

Your `F_in=0.2015` is close to your `F_out=0.9790` — in fact `F_out` is way above `F_in`, meaning your worst-case known sample and closest-case zero-day sample are essentially indistinguishable in fidelity space (or worse — zero-day fidelities can be *higher*). There's no geometric separation between the worst extremes of the two classes at all — `Δ` is deeply negative, `ε*` is negative, meaning no radius, however small, is certified safe.

But your accepted example (`F_max=0.8536`) shows that a *typical, non-adversarial* known sample sits nowhere near that worst case — its score `s(x)=1-0.8536=0.1464` is comfortably below `q=0.4055`. Conformal calibration doesn't need every known sample to be far from every zero-day sample; it only needs the observed *distribution* of nonconformity scores among known samples to be well-behaved enough that a data-derived quantile controls the *average* false-alarm rate. A few pathological samples (the ones dragging `F_in` down to 0.20) can sit anywhere in score-space without breaking Prop 3, because Prop 3's guarantee is about rank/quantile behavior of the *bulk*, not about the extremes.

In one sentence: conformal calibration is asking "does the threshold work well *on average*, using the data's own distribution as ground truth," while Proposition 2 is asking "does the threshold work *for every possible input*, including the worst one an adversary could construct" — and a system can be well-behaved in the first sense while having essentially no worst-case robustness margin in the second sense. These are not in tension with each other; they were never claiming the same thing.

Certified radius geometry, and why negative ε

The geometric picture

Think of $d_{min}(x)$ — trace distance from a state to its nearest prototype — as a literal distance in the space of density matrices. Around the prototype ρ_c, draw a ball of radius $d_{in} = \sqrt{1-F_{in}^2}$: every known sample's post-channel state lies inside this ball (that's assumption A2, translated via Fuchs–van de Graaf). Around every zero-day sample, mirror this: no known sample can be closer than $d_{out} = 1 - F_{out}$ to *any* prototype.

An adversarial perturbation δ with $\|\delta\|_2 \leq \varepsilon$ moves the encoded state by at most $L_\varphi \cdot \varepsilon$ (Lipschitz bound), which the depolarizing channel then contracts to at most $(1-p)L_\varphi \cdot \varepsilon$ (Prop 1). So the perturbed state can drift at most $(1-p)L_\varphi \cdot \varepsilon$ in trace distance. The certified radius R (Algorithm 3) is exactly the input-space radius over which this drift is guaranteed to stay inside a "safe zone" — small enough that the perturbed known sample never crosses into the ambiguous or zero-day-like region.

Positive Δ / positive radius

When $d_{in} < d_{out}$ (i.e. $\Delta > 0$), there's a genuine buffer zone between "known-sample territory" and "zero-day territory" — a moat of width Δ in trace-distance terms. As long as the perturbation-induced drift $2(1-p)L_\varphi \cdot \varepsilon$ stays smaller than that moat, both a threshold τ exists in the interval given in the theory doc, and a well-defined non-zero certified radius $R > 0$ exists: the sample can move around inside its own safe ball without ever being pushed to look like a zero-day.

Negative Δ / negative "radius"

When $d_{in} \geq d_{out}$, there is no moat — in the worst case, the known-sample ball and the zero-day boundary already overlap *before any perturbation is applied at all*. The formula for ε^* still computes a number, but that number is negative, meaning "even a perturbation of size zero already sits outside any provable safe region for the worst-case pair of points." A negative radius is not a physically meaningful radius (you can't have a ball of negative size) — it's the algebra signaling that the *precondition* for the certificate (A5) failed, not that the detector is malfunctioning.

Why this doesn't mean the algorithm is broken

This is the single most important distinction to carry into your writeup: **the certificate and the detector are two different objects.** The detector (Algorithm 3: encode → measure fidelity → conformal test → accept/reject) runs perfectly well regardless of whether Δ is positive or negative — it made a correct decision on your example sample (`F_max=0.8536` accepted). What fails when Δ ≤ 0 is only the *proof that this decision is guaranteed to survive an adversarial perturbation up to some radius*. No certificate existing is a statement about the limits of what can be *proven*, not a statement about what the detector actually does on real, non-adversarial (or even most adversarial) inputs — which is exactly what your Day 19 empirical robustness curve already showed: high acceptance rates held up to a real breakpoint around ε≈0.42, far beyond where the (negative) certified bound would suggest anything at all. The certificate is a conservative, sufficient (not necessary) condition — its absence is silence, not a negative result about the detector's actual behavior.